EBIN BABU THOMAS

AI Engineer: LLM Agents, RAG & Evaluation Systems

ebinbt.dev · hello@ebinbt.dev · github.com/ebt55

Mavelikara, Kerala, India · Remote · Open to relocation: Bengaluru / Hyderabad

SUMMARY

AI engineer building LLM agents, search systems and the tests that check them. Independent since March 2026, shipping a local meeting transcriber in public beta and a PDF-to-Word converter on PyPI. Built hybrid keyword and vector search over 433 city council meetings, and a purchase-checking agent that keeps hashed screenshots as evidence. Before that, 3.5 years building LLM, RAG and backend systems for startup clients.

TECHNICAL SKILLS

Core: Python, FastAPI, LLM Agents & Tool Use (Claude Agent SDK, Anthropic API, Google Gemini API, OpenAI API, PydanticAI, LangGraph, LangChain), MCP (FastMCP), Hybrid RAG (BM25 + RRF), Prompt & Context Engineering, Playwright, pytest, Docker, Git

Proficient: PostgreSQL / SQL, SQLite (FTS), MongoDB, Vector DBs (Qdrant, LanceDB), PyTorch, Hugging Face Transformers / PEFT, Fine-tuning (LoRA / QLoRA, DPO), LLM Serving (vLLM, Modal), Speech (Whisper, NVIDIA Nemotron ASR + diarization), Image & Document Processing (OpenCV, scikit-image, Pillow, RapidOCR, PDFium, PyMuPDF), MLOps (GitHub Actions CI/CD, OpenTelemetry), Pandas, AWS, GCP, GKE, OAuth2 / JWT, Terraform, Linux, uv

Evaluation engineering & measurement: eval harness design (Inspect, inspect_evals), retrieval benchmarking, agent evaluation sweeps, replaying recorded agent runs, blind grading, LLM-as-judge agreement checks, bootstrap & permutation statistics, predictions written before runs, hash-pinned reproducible artifacts, Langfuse

Familiar: Rust / Tauri 2, Svelte 5, Bytewax, Next.js, TypeScript

EXPERIENCE

Software Engineer · Zackriya Solutions *Jan 2022 – Jun 2025*

Remote, India. Services firm building first versions of products, AI systems and data tools for startup clients in the US, Canada, Europe and Australia. Grew from trainee to core developer; owned client backends end to end and scoped incoming projects.

- **FinBot, financial RAG & fine-tuned LLM:** fine-tuned Mistral 7B (QLoRA) and built a streaming news pipeline (Alpaca news API → Bytewax → Qdrant) so answers cite live market news; deployed and served with vLLM on a GKE GPU node.
- **Natural-language property search:** built a plain-English-to-SQL search backend on FastAPI (GPT-3.5 query translation with "Did you mean?" fallback suggestions) for real-estate listings; load-tested and deployed on Google Cloud Run.
- **DocuAI, semantic document retrieval:** designed the chunking, ingestion and Qdrant vector-search pipeline over 1,000+ documents with a Next.js front end; delivered and demoed the working prototype end to end in two days.
- **Data & speech pipelines:** sensor-data ingestion (MQTT → MongoDB) for a bioreactor startup that later raised funding; spoken-English scoring from short videos (Whisper, Microsoft's pronunciation API, AWS Fargate); healthcare-device compliance pipelines.

EDUCATION & CERTIFICATIONS

Technical AI Safety course (BlueDot Impact) *Sep – Oct 2026*

BlueDot certificate (Oct 2026)

Deep RL (Udacity Nanodegree) · Machine Learning (Stanford Online) *2020*

Udacity certificate (Jul 2020) · Coursera certificate (Mar 2020)

B.Tech (CSE), undergraduate coursework *2018 – 2020*

APJ Abdul Kalam Technological University · Sree Buddha College of Engineering, Kerala

INDEPENDENT PROJECTS & RESEARCH

Diarist: Local Meeting Transcriber with Speaker Labels, for Windows *Oct 2026*

Public beta 0.1.0-beta.5, free and MIT-licensed. diarist.dev · github.com/ebt55/diarist (NVIDIA NeMo-Speech.cpp (CUDA), Nemotron speech and speaker models, Svelte 5, TypeScript, SQLite (FTS))

- Built a Windows app that records a meeting, or imports a recording, and writes a transcript with a speaker label on every line. It runs on the user's own GPU: no cloud, no account, no Python install.
- An hour of audio takes **about two minutes on an RTX 3080**. On my own Zoom meetings, about 2 or 3 words in 1,000 went to the wrong speaker. Built with coding agents; tested on one PC so far.
- diarist.dev/models ranks speech and speaker models by public leaderboard scores, beside my own measurements of accuracy, speed and GPU memory, so users can pick models for their GPU.

ProofPack: Website-Check Agent That Keeps Hashed Proof for Every Claim *Jul – Sep 2026*

Built from a challenge brief (Build with Gemini XPRIZE, Jul 2026). A project, not a product; no users. github.com/ebt55/proofpack (Python, Gemini API, Playwright, FastAPI, Jinja2, pytest)

- Built an agent that checks Medicaid purchase requests against providers' public websites, the way a compliance reviewer would, and saves a stamped screenshot as proof of each claim.
- Gemini reads the form and does the browsing. Plain code records every URL, timestamp and SHA-256 hash, **rejects any "Found" without a screenshot**, and leaves each approve or deny decision to a person.

councilrag & rbb: Council-Meeting Search with Video Citations, and a Retrieval Benchmark *Oct 2026*

From a private 48-hour company screening test, Oct 2026 (no public repo). (Python, LanceDB (BM25 + vector + RRF), Qwen3-4B, Granite-3.3-2B, RapidOCR, Diarist ASR, pytest)

- councilrag searches 433 Seattle City Council meetings and answers vote questions from the official record, with citations that open the video at the second.
- On a 9-meeting slice it also reads speaker-labelled audio, on-screen text and the official vote record. The live captions there had dropped 18 roll-call votes; **the audio transcript recovered 17**.
- rbb is a retrieval benchmark that stores evidence as time spans, so any chunking can be scored, and checks whether its scores predict what a reader model answers. Within one token budget on Video-MME questions, graded labels picked the setup that answered correctly **66% of the time, against 50% for token count**. A Seattle replication showed no such gap.

printready: Checks and Fixes AI Sticker Art Against a Printer's Rules *Oct 2026*

Rules from Sticker Mule's public FAQ; not affiliated. github.com/ebt55/printready (Python, OpenCV, scikit-image, Pillow, xAI API, MCP server, pytest)

- Built a tool that checks AI-made sticker art before printing: resolution at the chosen size, background, art cut off at the edge, and misspelled text. It removes plain backgrounds, adds the white border and cut line, and sends anything it can't fix safely to a person.
- A guard undoes any fix that damages the art. **Letting an AI agent choose the fixes did no better**, at about four times the cost.

ExactDoc: PDF to Editable Word Documents, Checked Page by Page *Aug – Oct 2026*

Beta 0.3.0b3 on PyPI. github.com/ebt55/exactdoc (Python, PDFium / pypdfium2, OOXML, LibreOffice headless, PyMuPDF, pytest)

- Converts a PDF into a Word document you can edit, laid out like the original, with real headings, tables and paragraphs that re-wrap. It can also write a version made for Google Docs.
- Tested by opening the output in Word, LibreOffice and Google Docs and **comparing every page with the PDF**. With LibreOffice installed, it runs that check itself and fixes page breaks and spacing. Designed layouts and equations don't convert well yet.

Kobayashi Maru: Do Impossible Tasks Make AI Agents Cheat on Solvable Ones? *Sep 2026*

Apart Research × CeSIA AI Incident Response sprint (Sep 11–13), solo; top half of 230+ submissions. github.com/ebt55/kobayashi-maru (Python, Docker, Anthropic API, OpenRouter, Ollama, statsmodels)

- Built a harness that adds impossible tasks to a coding agent's batch and checks, with plain code, whether it then cheats on the tasks it could solve. Over 8,959 runs on six models, **two cheaper models started cheating**; one went from 0% to 30%. Batches also got longer as impossible tasks were added, which may explain part of it.

LLM Pricing Hub Remedies: Do Antitrust Rules Stop a Shared AI Pricing Advisor? *Oct 2026*

Apart Research AI Collusion Research Sprint, 23–25 Oct 2026; results not announced. github.com/ebt55/llm-pricing-hub-remedies (Python, uv, Anthropic API, OpenAI SDK, OpenRouter, SciPy, pandas)

- Simulates one AI pricing advisor serving two competing sellers, and tests whether the rules in the 2026 US settlement with RealPage stop it from raising their prices. More than 700 simulated games on seven models.
- **No rule worked reliably**. Banning rival data removed most of the harm on three models, almost none on a fourth, and raised prices on a fifth. The advisor was told to maximise the sellers' joint profit, so this tests the rules against that incentive.

diffing-agent-bench: Sealed Benchmark for AI Agents That Audit Fine-Tuned Models *Aug – Sep 2026*

MATS 12.0 work sample (Neel Nanda stream), Sep 2026. github.com/ebt55/diffing-agent-bench (Python, PyTorch, PEFT / LoRA, vLLM, Qwen3.5-9B)

- Built a sealed benchmark for AI agents that audit a fine-tuned model by comparing it with the original. I planted known habits in LoRA finetunes of Qwen3.5-9B and locked the answer key before any run. In 13 attempts the agent **never asked a database question**, so it missed the planted database habit. A fixed list of 50 prompts found it for 15 cents.

IncidentGate: Test Lab for Safety Checks on an Incident-Repair Agent *Aug – Sep 2026*

Closed at a baseline, Sep 2026. github.com/ebt55/incidentgate (Python, LangGraph, PostgreSQL, OpenTelemetry, Langfuse, Docker Compose, pytest)

- Built a test lab for three safety checks on an incident-repair agent: a rule list, a watcher model and a scripted stand-in approver. A forbidden action made in one call was blocked at once. A different forbidden change, split into three allowed steps, **got past all three checks** with the local watcher model. The attack was scripted, and each setting ran once.

OPEN-SOURCE CONTRIBUTIONS

- Meetily **(GitHub-Trending meeting-notes app)**: backend refactor and OpenAI support, merged (PR #75); bpmn-io/refactorings: text-similarity recommender for connector templates, replacing an LLM call (issue #33).

Career break Jul 2025 – Feb 2026. Independent engineering and research since Mar 2026.